



Bank Loan Analysis – Business Requirements & Insights Document



Business Objective

The objective of this project is to analyze and monitor the bank's lending activities and loan portfolio performance.

This report enables tracking of key loan metrics, assessment of credit risk, and identification of trends to support data-driven lending decisions.



Scope of Analysis

In Scope:

- Loan applications, funded amounts, repayments, and loan status
- Borrower profiles and credit risk indicators

Out of Scope:

- External financial market data
 - Non-loan banking products
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Key KPIs

- Total Loan Applications
 - Total Funded Amount
 - Total Amount Received
 - Average Interest Rate
 - Average Debt-to-Income Ratio (DTI)
 - Month-to-Date (MTD) & Month-over-Month (MoM) Metrics
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Good Loan vs Bad Loan Analysis

Good Loans (Low Risk):

- Loan Status: *Fully Paid, Current*
- Metrics:
 - Good Loan %

- Total Applications
- Funded Amount
- Amount Received

Bad Loans (High Risk):

- Loan Status: *Charged Off*
- Metrics:
 - Bad Loan %
 - Total Applications
 - Funded Amount
 - Amount Received

👉 **Objective:** Evaluate portfolio quality and identify risk exposure

Dashboard Insights

1. Loan Performance Overview

- Tracks applications, disbursements, and repayments
 - Helps assess overall portfolio health and growth trends
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2. Monthly Trends Analysis

- Visualizes loan applications, funded amount, and repayments over time
- 👉 **Insight:** Identifies seasonality and growth patterns in lending
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3. Regional Analysis

- Loan distribution across states
- 👉 **Insight:** Detects high-performing and high-risk regions
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4. Loan Term Analysis

- Distribution of loans across different terms (36, 60 months)
- 👉 **Insight:** Helps understand borrower preferences
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5. Employment Length Analysis

- Loan metrics based on borrower employment history

👉 **Insight:** Longer employment = lower default risk

6. Loan Purpose Analysis

- Breakdown by purpose (debt consolidation, credit card, etc.)

👉 **Insight:** Identifies major borrowing needs

7. Home Ownership Analysis

- Loan distribution by ownership type (own, rent, mortgage)

👉 **Insight:** Indicates borrower stability and risk

Details Dashboard

Provides a consolidated view of:

- Loan-level data
- Borrower information
- Loan status and repayment details

👉 **Objective:** Enable detailed exploration and quick decision-making

Domain Understanding

Bank loans help individuals and businesses meet financial needs.

Banks collect and analyze loan data from:

- Applications
- Credit reports
- Internal systems
- Online portals
- Third-party sources

👉 This data supports **risk assessment, decision-making, and portfolio management**

Loan Approval Process

- Application Submission

- Verification (Identity, Income, Employment)
 - Credit Check
 - DTI Evaluation
 - Risk Assessment
 - Approval / Rejection
 - Disbursement
 - Repayment & Monitoring
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Key Business Use Cases

- Credit Risk Assessment
 - Loan Approval Decision-Making
 - Portfolio Performance Monitoring
 - Fraud Detection
 - Customer Segmentation
 - Profitability Analysis
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Data Fields & Importance

Key fields used:

- Loan ID → Unique identifier
- Loan Status → Loan health tracking
- Annual Income → Repayment capacity
- DTI → Risk indicator
- Interest Rate → Loan cost
- Loan Amount → Principal value
- Employment Length → Stability indicator
- Home Ownership → Financial stability
- Purpose → Borrowing intent

 These fields help in **risk analysis, segmentation, and decision-making**

Final Insights & Recommendations

- Monitor **high-risk (bad loans)** to reduce defaults
 - Focus on borrowers with **low DTI & stable employment**
 - Optimize lending strategies based on **region & loan purpose**
 - Improve portfolio health through **risk-based segmentation**
 - Use dashboards for **continuous monitoring & decision support**
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Conclusion

This project provides a comprehensive view of loan performance, borrower behavior, and credit risk. It enables banks to make informed decisions, reduce risk, and improve overall portfolio efficiency.